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HYPNEXIS '26

Team Details

Problem Statement: AI-Powered Real-Time Detection and Prevention of Voice-Cloning Impersonation Attacks and Scam Calls

Theme: Artificial Intelligence, Cybersecurity and Fraud Prevention

Team Name: PyRates Of Hackathon

Team Members: Tanya Sriram, Varada Sharma, Pranav Singh, Kinshuk Mohul

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ABOUT THE PROJECT



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THE PROBLEM

Every day, scam callers pose as banks, courier companies, or the police, pushing people to share an OTP or move money within minutes. Voice-cloning tools now let scammers sound like anyone — a relative, an official, a trusted contact. The person on the call has only seconds to decide, and no tool to help them while it's happening.

THE SOLUTION-VoiceGuard

VoiceGuard listens to a phone call — either live or an uploaded recording — and checks two things at the same time:

1. Is this a scam?
2. Is this a real human voice?

PRIVACY BY DESIGN:
nothing is analysed until the user ticks a consent box. Audio is processed in memory only and is never written to disk or kept after the session.

RESULTS

NOT A SCAM CALL
Nothing said matches scam tactics and the voice sounds natural.

AI-GENERATED VOICE, LIKELY NOT A SCAM
The voice sounds synthetic, but the words are harmless.

SCAM CALL DETECTED
The caller is using known scam tactics, with or without an AI voice

How a person uses it:

Live, by microphone: put the call on speaker next to the laptop.
Live, by tab audio: share the browser tab of an internet call.
Afterwards: upload a call recording

What the person sees:

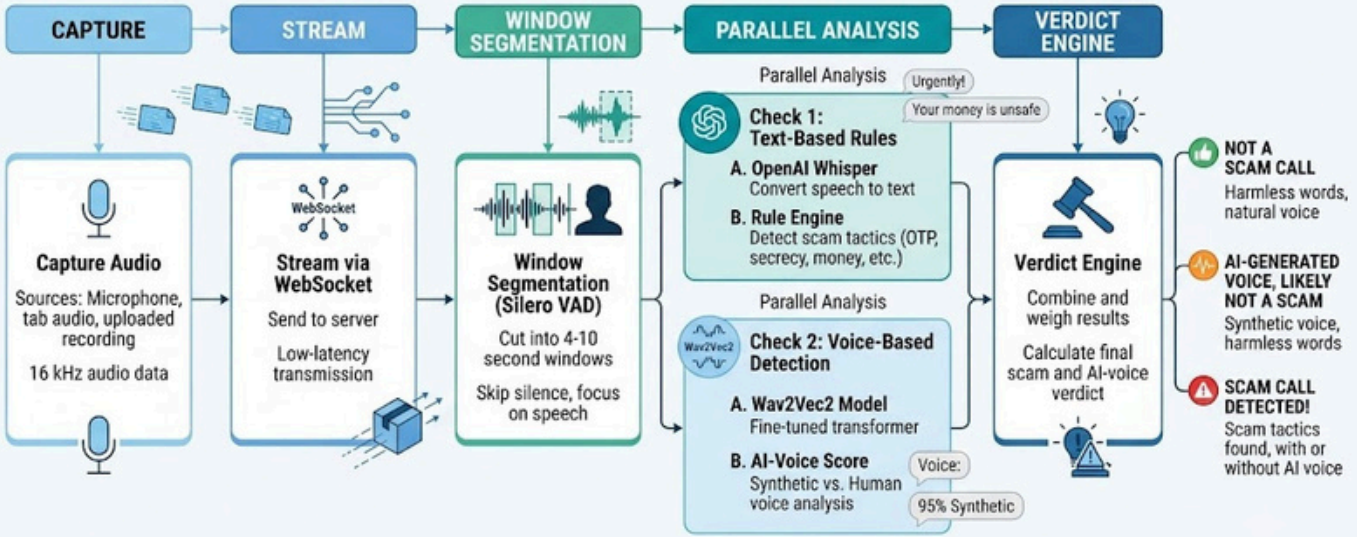
- The result, updated every few seconds during the call.
- A live transcript, with risky lines highlighted. The scam tactics that were matched, and a voice score.
- Advice on what to do next, with the helpline number 1930.

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TECHNICAL APPROACH

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PROCESS FLOW ARCHITECTURE



Privacy by Design: User consent required before analysis. Audio processed in memory.

How the voice model was trained

1 Base model
A public Wav2Vec2 anti-deepfake detector from NII's Yamagishi Lab, about 95 million parameters.

2 Training data
20.5 hours of speech: 13.4 hours of AI voices and 7.1 hours of real people, in Hindi, English and Hinglish.

3 AI-voice samples
Generated with TTS engines (Microsoft neural voices, Google TTS, Kokoro, Piper) plus public ElevenLabs, OpenAI and Gemini voice sets.

4 Real-call conditions
Every clip is also heard over a phone line, through a speaker and room into a microphone, in noise, and through MP3, AAC, Opus and AMR codecs.

5 Fine-tuning
4,000 training steps on the GPU. The result is averaged 50/50 with the base weights, which keeps its reach on voices it has not heard.

6 Export
Converted to ONNX with 8-bit weights: 120 MB on disk, small enough for a free server.

TECHNOLOGY STACK

Languages	Python 3.11, JavaScript (ES modules), HTML5, CSS3
Backend	FastAPI and Uvicorn; WebSocket for live audio, REST for uploads; Pydantic data models
Speech-to-text	OpenAI Whisper, run locally with faster-whisper (CTranslate2, 8-bit)
AI-voice detection	Wav2Vec2 transformer, fine-tuned for this project; ONNX Runtime with 8-bit weights
Audio	Web Audio API and AudioWorklet in the browser; Silero VAD, FFmpeg (PyAV) and NumPy on the server
Frontend	Plain HTML, CSS and JavaScript with no framework; animated canvas visualisation
Training	PyTorch on an NVIDIA RTX 5060 GPU, with a custom audio augmentation module
Testing	pytest: 156 automated tests covering the API, the WebSocket protocol, the verdict logic and the rules
Deployment	Docker image on Render (backend), Vercel (frontend), GitHub for source control

Github: https://github.com/Reactive07-Pranav/voiceguard
Prototype: https://voiceguard-frontend-sage.vercel.app/

FEASIBILITY AND VIABILITY

FEASIBILITY

VoiceGuard is already deployed and working end to end, and it costs nothing to run.

- **Zero cost: the backend runs on a free 512 MB server and the frontend on a free static host.**
- **No paid services: the AI models run on our own server, so there are no API keys or per-call fees.**
- **Nothing to install: it works in a normal browser on a laptop.**
- **Fits in memory: the server starts at 286 MB and peaked at 406 MB in testing.**
- **Tested: 156 automated tests pass.**

VIABILITY

Known limits and the way forward

LIMIT	FIX
The free server is slow(about 9 seconds per window)and sleeps when idle	Paid server removes both limits
Whisper accuracy	A larger server can run a more accurate Whisper model with one setting changed.
Loudspeaker audio is harder to judge than uploads	Prefer uploaded recordings

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IMPACT AND BENEFITS

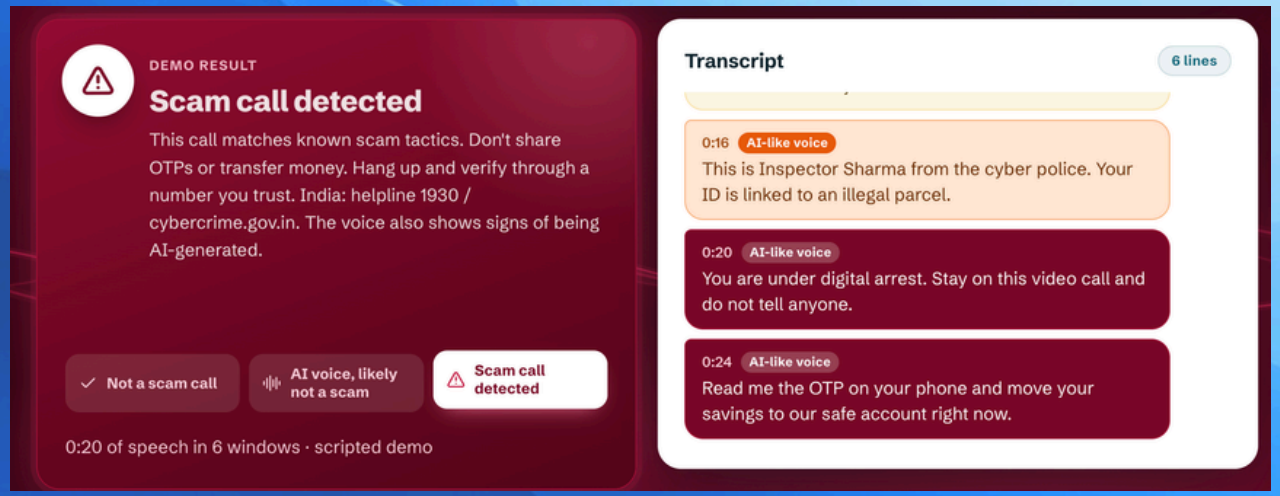
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IMPACT

The warning arrives during the call, before an OTP is shared or money is sent, which is the only moment it can prevent the loss.

- **Without VoiceGuard: the victim hears pressure and a familiar voice, and decides alone, in seconds.**
- **With VoiceGuard: the victim sees a red warning, the matched tactics and plain advice, and decides with evidence.**





BENEFITS

- 🛡️ **Protects people under pressure:** a clear red result and plain advice counter the urgency a scammer creates.
- 📣 **Exposes cloned voices:** flags an AI voice even when the caller sounds like a relative or an official.
- 💬 **Built for Indian calls:** works with **Hindi, English and mixed Hinglish speech.**
- 🔎 **Explains itself:** shows the transcript and the exact tactics matched, so the user can judge the evidence.

Measured results

250 of 255	2 of 119	99 to 168
AI-voice calls caught on recordings as made	real calls wrongly flagged on recordings as made	AI calls caught through a speaker and microphone, before and after fine-tuning

Audio condition	AI calls caught	Before fine-tuning	Real calls wrongly flagged
Recording as made	**250 of 255**	244	2 of 119
Phone line (simulated)	**239 of 255**	223	3 of 120
Speaker to microphone (simulated)	**168 of 255**	99	4 of 120

RESEARCH AND REFERENCES

- **AntiDeepfake (NII Yamagishi Lab):** W. Ge, X. Wang et al., "Post-training for Deepfake Speech Detection", arXiv:2506.21090. The starting point for our voice model (licence: CC BY-NC-SA 4.0, non-commercial).
- **wav2vec 2.0:** A. Baevski et al., 2020. The self-supervised speech model the detector is built on.
- **Whisper:** A. Radford et al., OpenAI, 2022. Multilingual speech recognition, run with faster-whisper.
- **Silero VAD:** voice activity detection, used to skip silence and judge only real speech.
- **Speech corpora: FLEURS, LibriSpeech, MInDS-14** and **NPTEL** lectures for real voices.
- **Scam patterns:** public advisories from **India's National Cyber Crime Reporting Portal** (cybercrime.gov.in) and **helpline 1930.**
- **AI & Logic: OpenAI Whisper** (speech-to-text) ,**Wav2Vec2 anti-deepfake model, fine-tuned on 20.5 h of Hindi, English and Hinglish speech** (AI-voice detection), **Weighted rule engine** (scam tactics).
- **Development: Claude** (Anthropic) as AI pair-programmer for architecture, code, the model training pipeline, optimisation and bug fixing
- **Deployment & Presentation: Vercel (Edge Hosting) | Keynote & Canva (Deck Design)**